Experiment plan: open items

2026-11-01

This is the list of changes the experiment plan in `paper.tex` needs before Stage 3 starts, from a review of `paper.tex` and `methodology.tex` on 2026-11-01. Each item names the problem, where it shows, and the fix. Items are ordered by how much they change what gets run. The ones in §2 change the runs; the ones in §3 change only the documents. Every item is OPEN until ticked off here.

Terms used below. An *arm* is one configuration in the comparison (A to F in the paper's ablation table). A *leaf* is one scored trajectory. *Fast success* means the final run succeeds within 1.25× the time-optimal batch time. The *probe rate* is the fraction of first replies that are probe code rather than a controller. The *blind reference* is the best hand-written controller that does not know the hidden limit; the *oracle* is the one that does.

1 What holds up

Not to be changed.

- The gating logic. Stage 0 proves the hidden limit is worth knowing before any model runs; Stage 0b proves the runner carries the solution; Stages 1 and 2 set the floor and the training target; Stage 3 checks the plumbing against a known ceiling. Nothing downstream rests on an unverified step.
- The ablation isolates the claim. E against D tests the rollout tree; F against E tests the per-turn baseline; the trained arms match on leaves, and both equal-rollout and equal-wall-clock comparisons are reported.
- Fast success as the primary metric. Raw success is saturated by any slow cautious controller.
- The environment. The limit leaves no trace until it is crossed, so a probe is the only way to learn it, and Stage 0 showed the gap between blind and oracle is wide.
- The immediate next step. Re-run Stages 1 and 2 under the optional probe with the batch record and 1000-token turns, then decide on Stage 3. Items 1 to 4 should be settled before Stage 3 starts, since they change what is run and how many seeds it needs.

2 Items that change what gets run

1. **The statistical test cannot reach significance at three seeds.** OPEN
   Problem. The evaluation section specifies a paired Wilcoxon signed-rank test over per-seed fast-success rates with at least three seeds per arm. The test ranks the per-seed differences between two arms; with n pairs the smallest possible two-sided p is $2/2^n$, so three seeds cannot get below 0.25.

Seeds per arm	Smallest two-sided p
3	0.25
4	0.125
5	0.0625
6	0.031

Fix. One of: (a) six seeds per arm for the arms in the headline comparison (D and F at least); (b) pair over seed-by-batch units, so the test has seeds times held-out batches pairs; (c) drop the p-value and report the effect size with a bootstrap interval over seeds. Whichever is chosen, write it into § Evaluating in `paper.tex` before the training runs are scheduled, because it sets the seed count.

2. **The token budget is inconsistent across arms.** OPEN

Problem. The same quantity is set to four different values.

Where	Value	Applies to
`paper.tex`, Model and training paragraph	400 per turn	stated for all training
`paper.tex`, Appendix memory table	$T = 400$	memory arithmetic
`configs/loop_grpo_crystallizer.yaml`	400	arm C (single-turn GRPO)
`configs/loop_grpo_branched_crystallizer.yaml`	1000 per turn	arms E, F
`scripts/05_train_grpo_branched.py` default	400	if the config omits it
Stage 1 and 2 as run	600	frozen arms A, B
Stage 1 and 2 re-run (planned)	1000	frozen arms A, B

Stage 1 lost 28 of 96 controllers to the cut at 600 tokens and Stage 2 lost 44 of 96 probes. Arm C at 400 would be handicapped against arm A at 1000 and could fail the plumbing check for the wrong reason.

Fix. 1000 tokens per turn everywhere: the single-turn config, the script default, the paper's paragraph, and the memory table (redo the arithmetic at $T = 1000$ or state that the table is illustrative at 400). Arm D's config, when it is written, inherits the same value.

3. **No probe-rate gate before Stages 4 to 6.** OPEN

Problem. The probe is optional. If the frozen model under the re-run protocol almost never probes, arms D, E and F collapse to arm C and the rollout tree carries no turn-1 signal to learn from. With $G_1 = 4$ and a probe rate of ten percent, about a third of prompts have at least one probing variant; at one percent, about four percent do. The extensions section already names ten percent as the threshold for a prompt variant to be trainable, but Stage 2's *Decides* clause does not use it, and it also does not say what happens if arm B comes out below arm A.

Fix. Add to Stage 2: a probe rate under the optional protocol of at least ten percent is the gate for Stages 4 to 6; below it, the brief is changed (not the training) and Stage 2 is re-run. Add the reading for B below A: the probe run hurts, so the cause (budget, crashes, invented results) is fixed in the brief before training.

4. **Checkpoint selection on the test set.** OPEN

Problem. Results are "taken at the best checkpoint" and reported on the 12 held-out batches. If the best checkpoint is chosen on those same batches, the reported number is optimistically biased, and more so for arms with noisier curves.

Fix. Choose the checkpoint on training reward or on a separate validation draw of batches, then evaluate the held-out set once. Write the rule into § Evaluating.

5. **The evaluation set is thin.** OPEN

Problem. Fast success depends strongly on where the limit sits in $[0.08, 0.25]$: at the bottom the blind controller is already fast, at the top the gap is largest. Twelve batches cover that range coarsely, and the per-batch rate is itself a sampled quantity at temperature 0.8.
Fix. 36 to 48 held-out batches with limits stratified across the range (equal counts per tercile at least). Evaluation is inference only, so this is cheap. Report fast success per tercile as well as overall.

6. **Add a frozen arm that is told the true limit.** OPEN
Problem. If arm F fails, nothing in the current design says whether the model cannot write the ride-below-limit controller or cannot probe. The Stage 2 record at 600 tokens already shows the controller itself failing: setpoints of 0.18 to 0.20 with no margin, 24 of 96 batches ruined after a median of 20 steps.
Fix. One inference-only arm: the frozen model, single turn, with the true $\sigma_{\max}$ written into the brief. It is the model analogue of Stage 0's informed side. Its fast-success rate is the ceiling any probe-aware arm can reach with this model's controller-writing, and subtracting it from the oracle gives the controller gap; subtracting arm B from it gives the probe gap. Add it to the ablation table as A^+ and to Stage 1.

7. **Arms A to D form a two-by-two, not a chain.** OPEN
Problem. The factors are frozen versus trained and one turn versus two. The ablation table's "what it isolates over the previous arm" entry for arm C compares it to arm B, which changes both factors at once.
Fix. Present A, B, C, D as a grid. C against A is the effect of training without a probe; D against B is the effect of training with one; B against A and D against C are the effect of the probe, frozen and trained. E and F then extend D. Rewrite the table's last column accordingly.

8. **Stages 4, 5 and 6 are one experiment, not three rungs.** OPEN
Problem. No stop condition is stated between Stage 4 and Stage 5 or between 5 and 6, and none makes sense: the question is the comparison, and the arms share seeds. Only Stage 3 is a real gate, and it has no numeric threshold, unlike Stages 0 and 0b.
Fix. Give Stage 3 a gate: arm C reaches the blind reference (mean reward about +27, fast success about 4 to 7.5 percent) within the step budget on at least two of three seeds; failing that, the reward or the plumbing is wrong and nothing else runs. Merge Stages 4 to 6 into one stage with three arms run together on the same seeds, and keep the *Decides* text of the current Stage 6 as its decision rule.

9. **The stress set contradicts the brief.** OPEN
Problem. The brief tells the model the limit lies between 0.08 and 0.25. A stress set with limits outside that range therefore tests what the model does when its stated prior is wrong, which is a different question from robustness.
Fix. Either widen the range stated in the brief so the stress set lies inside it, or keep the stress set and say in § Evaluating that it measures behaviour under a wrong prior. Either way, drop the word "planned" or schedule it.

3 Consistency fixes in the documents

10. **A stale mechanism check.** OPEN
Problem. The first diagnostic in § Evaluating says that when turn 1 emits a policy its reward should improve faster under F. That describes the old protocol, where a turn-1 policy was rehearsed and its reward fed back. Under the current protocol a turn-1 controller ends the

task.
Fix. Replace it with: the probe rate over training, and the reward of probed against unprobed trajectories. Under F the probed trajectories' turn-1 advantage should become positive before the probe rate rises; under D it cannot be separated.

11. **The tree weights a turn-1 controller as a full branch, and the paper does not say so.** OPEN
Problem. In `src/branched_trainer.py` (the block commented "The model wrote the controller in turn 1"), a first reply that is a controller is entered into the reward grid G_2 times, so $\widehat{V}_0$ weights it like a probing variant and its turn-1 tokens appear G_2 times in the loss. This is a reasonable choice, but it zeroes the within-prefix variance for those variants, which inflates the between/within signal ratio diagnostic whenever some variants probe and others do not.
Fix. State the choice in the Branched rollout subsection. Compute the signal ratio over probing variants only, or report it alongside the probe rate so the reader can tell the two apart.

12. **Matched leaves is not matched first-turn samples.** OPEN
Problem. Arm D draws 16 independent first turns per prompt; arms E and F draw 4. That is exactly what E against D is meant to expose, but the table caption says only that leaf counts match.
Fix. One sentence in the caption: D has G independent first-turn samples, E and F have G_1, and the E–D comparison is the price of that reduction against the saving in generate calls.

13. `methodology.tex` **describes the old protocol.** OPEN
Problem. It has typed `policy`/`main` fences, a rehearsal turn with the reward fed back, and an error floor of -200. `paper.tex` has content-defined replies, an optional probe, and a floor of -300. The two contradict each other.
Fix. Retire `methodology.tex`, or add a line under its title saying it is superseded by `paper.tex` as of 2026-10-26 and is kept for the derivation of the per-turn loss decomposition only.

4 Order of work

1. Items 2 and 3, because the Stage 1 and 2 re-runs need the 1000-token budget and the probe-rate gate is read off their result.
2. Items 1, 4 and 5, because they fix the seed count, the checkpoint rule and the evaluation set before any training run is scheduled.
3. Item 6, inference only, alongside the Stage 1 re-run.
4. Items 7, 8 and 9, text changes to the plan, before Stage 3.
5. Items 10 to 13, document consistency, at any time.